ARES V3: Deterministic Static Analysis for Solana Smart Contracts via Multi-Phase Taint Tracking and Macro-Aware AST Parsing

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Abstract

Existing tools for Solana smart contract security fall into two camps: generic LLM scanners with low detection rates and astronomical false-positive rates, or closed-source SaaS platforms that cost money, lock your code in the cloud, and cannot be reproduced. We present **ARES V3**, an open-source deterministic static analysis framework built specifically for the Solana ecosystem. It runs four phases in sequence: (1) fast regex heuristics, (2) macro-aware AST parsing with `syn` and `proc-macro2`, (3) intra-procedural taint tracking from untrusted sources to sensitive sinks, and (4) a **deterministic local judge** that suppresses systematic false positives using AST metadata alone — no calls to external LLM APIs.

We evaluate ARES V3 on a deliberately split benchmark. **Segment A** is 11 deterministic stubs (50–150 LOC each) that isolate single vulnerability classes for regression testing. **Segment B** is nine real production repositories (10K+ LOC each) previously audited by professional firms (Ackee, Code4rena, Neodyme, OtterSec, Kudelski, Trail of Bits). On Segment B, ARES V3 reaches **93% known-audit recall** (26/28 published findings) and **0.60 precision**, while keeping **100% detection** on Segment A. For comparison, the commercial state-of-the-art scanner Trident Arena scores 70% detection with a 26.56% false-positive rate. The entire pipeline runs locally in **<5 seconds per protocol** at **zero API cost**.

Our main contributions are: (a) a four-phase deterministic pipeline tailored to Solana that hits >90% recall on real audited code without LLMs or dynamic fuzzing, (b) an honest two-segment benchmark design that separates regression validation from real-world capability assessment with per-protocol recall capped at 100%, (c) a local judge algorithm that improves precision from 0.55 to 0.60 by suppressing systematic false positives using only AST metadata — no LLM calls, no non-determinism, and (d) macro-aware parsing for both Anchor `#[derive(Accounts)]` and Solitaire `#[derive(FromAccounts)]` that closes total detection gaps on Wormhole (0% → 100%) and Solend (0% → 100%). All code, ground-truth datasets, and benchmark harnesses are published openly for reproduction.

Keywords: smart contract security, Solana, static analysis, taint analysis, macro parsing, benchmark, false-positive suppression

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1. Introduction

Solana occupies a unique position in the blockchain space: theoretical throughput of 65,000 TPS and low fees have attracted billions of dollars in Total Value Locked. Yet Solana's technical design — explicit account models, Cross-Program Invocation (CPI) semantics, and heavy use of Rust macros in the Anchor and Solitaire frameworks — creates attack surfaces that generic analysis tools simply do not see.

History shows that critical bugs survive even professional audits:

- **Wormhole (February 2022):** A $325M signature-verification bypass hid inside macro-generated code from `#[derive(FromAccounts)]` in the Solitaire framework. The `instruction_acc: Info<'b>` field in `VerifySignatures` never generated a `Signer<>` check because the macro omitted it for that field [6].
- **Mango Markets (October 2022):** Oracle manipulation and forced liquidation caused ~$100M in losses. The bug required understanding how price-manipulation instructions interacted with liquidation instructions within the same program [7].
- **Solend (June 2022):** An oracle attack exploited unchecked numeric casts in raw `AccountInfo` handlers [8].

These bugs share a common thread: they live in Solana-specific semantics that regex scanners and generic LLMs miss.

Current tools fall into three categories, and each has a fatal weakness:

Generic LLM scanners (Claude Opus, GPT-4) reason well about general code but are blind to Solana runtime semantics. On the Trident Arena benchmark, Opus 4.6 detects only 11/30 critical/high findings (37%) and GPT-5.2 xhigh detects 10/30 (33%), with an 86.67% false-positive rate [1]. They cannot track data flow from a raw `AccountInfo` parameter into an `invoke()` call, nor do they understand that `seeds = [...]` in an Anchor macro constrains PDA ownership.

Commercial SaaS platforms like Trident Arena do better (21/30, 70%) but are closed boxes: no source code, no local CLI, no CI/CD hook, and opaque pricing [1]. Worse, their benchmark includes the Watt protocol, whose source code has never been published — only an Ackee audit PDF exists. No independent researcher can verify Trident Arena's claimed 7/11 detection on Watt. This is not open science; it is a marketing slide.

Open-source static analyzers such as cargo-audit, cargo-geiger, and Sec3 X-ray focus on dependency CVEs or compiled bytecode. They do not analyze instruction-handler logic, CPI

validation patterns, or PDA seed constraints — exactly the code paths that attackers exploit in Solana.

1.1 Research Questions

This work asks three concrete questions:

1. **RQ1:** Can deterministic static analysis — without LLMs or dynamic fuzzing — achieve >90% recall of published audit findings on real Solana production code?
2. **RQ2:** How should a benchmark for smart-contract security tools be designed to avoid "benchmark theater," where a scanner is overfitted to a small dataset and its scores do not translate to real-world capability?
3. **RQ3:** Can systematic false positives be suppressed deterministically using only local AST metadata, avoiding the cost, latency, and non-determinism of LLM-as-a-judge APIs?

1.2 Contributions

We make four contributions:

- **K1 — A four-phase Solana-specific pipeline:** We design and implement a deterministic static analysis pipeline (regex → AST → taint → local judge) that reaches 93% known-audit recall on nine production repositories without relying on LLMs or fuzzing.
- **K2 — An honest two-segment benchmark:** We introduce an explicit split between deterministic regression stubs (11 isolated vulnerability classes) and real-world capability assessment (9 production repos of 10K+ LOC each), capping per-protocol recall at 100% to prevent mathematical absurdity.
- **K3 — Deterministic local false-positive suppression:** We show that AST metadata already collected during parsing — typed Anchor field counts, CPI validation contexts, safe-wrapper arithmetic patterns — is sufficient to suppress systematic false positives deterministically, raising precision from 0.55 to 0.60 without hurting recall and without any external API cost.
- **K4 — Macro-aware parsing for Anchor and Solitaire:** We implement structural extraction of `#[derive(Accounts)]` and `#[derive(FromAccounts)]` that closes total detection gaps on Wormhole (0% → 100%) and Solend (0% → 100%), where regex-only scanners failed completely because macros hide the actual validation logic.

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2. Related Work

2.1 Static Analysis for Blockchain

EVM tools. Slither [9] is the best-known static analyzer for Solidity. It compiles Solidity to an intermediate representation and detects common patterns such as reentrancy, unchecked transfers, and uninitialized storage pointers. Slither reaches ~90% recall on certain EVM datasets, but its semantics are fundamentally different from Solana: EVM has no explicit account model with ownership and PDA seeds. The concepts do not map.

Solana tools. Sec3 X-ray [10] performs static analysis on compiled Solana bytecode, which requires the program to build successfully first and loses access to macro-level Rust source. cargo-audit [11] and cargo-geiger [12] check dependencies for known CVEs and `unsafe` Rust usage — useful, but irrelevant to program-specific logic bugs such as missing signer checks or unvalidated CPI targets.

2.2 Fuzzing for Solana

Trident [2] is a property-based fuzzing framework for Solana built around Trident SVM, a fast Solana transaction executor. It supports stateful fuzzing through "fuzzing flows" — randomized instruction sequences meant to explore rare execution paths. Trident Arena [1] layers a multi-agent AI system on top of Trident to produce audit reports.

Dynamic fuzzing has three hard limits on Solana: (a) it needs a manually written fuzz harness for every program, (b) coverage-guided fuzzing struggles with deep multi-instruction paths (e.g., deposit → oracle manipulation → liquidation), and (c) it cannot analyze code that fails to compile, which is common when cloning production repos with complex workspace dependencies. Static analysis complements fuzzing by finding bugs in code that does not yet build or run.

2.3 LLMs for Security Auditing

Anthropic's SCONE-bench [3] evaluated ten frontier models on 405 exploited EVM smart contracts. Success is defined economically: the agent must generate an exploit script that increases its native-token balance by ≥0.1 ETH in simulation. The best models exploited 51.11% of contracts. A follow-up zero-day evaluation on 2,849 fresh contracts found two novel zero-days worth $3,694 [3].

On Solana, LLMs fail for different reasons: they lack runtime semantics (account ownership, CPI seeds, discriminator checks), they hallucinate heavily on macro-heavy code (86.67% false positives on Trident Arena [1]), and API costs make routine scanning uneconomical ($1,738 per successful exploit on SCONE-bench [3]).

2.4 Benchmarking Security Tools

The DARPA Cyber Grand Challenge (2016) introduced open benchmarking for security analysis, but focused on binary exploitation. smart-bench [13] evaluates EVM static analyzers on historical vulnerability datasets, yet no Solana equivalent exists.

Trident Arena [1] uses a retrospective benchmark: six previously audited protocols with 30 critical/high findings as ground truth. The benchmark cannot be fully reproduced because Watt source is unpublished, and it lacks segmentation between regression testing and real-world assessment. Our two-segment design addresses both problems.

Benchmark Results (Retrospective). Table 1a presents the per-protocol detection rates reported by Ackee-Blockchain for Trident Arena, Claude Opus 4.6, and GPT-5.2 xhigh on the six audited protocols. Table 1b summarizes aggregate metrics [1].

Protocol	Critical/High Total	Trident Arena	Opus 4.6	GPT-5.2 xhigh
Axelar	7	5/7 (71%)	0/7	0/7
Bert Staking	2	1/2 (50%)	1/2	1/2
Dexalot	5	4/5 (80%)	2/5	2/5
Pump Science	2	1/2 (50%)	1/2	0/2
MetaDAO	3	3/3 (100%)	1/3	1/3
Watt	11	7/11 (64%)	6/11	6/11
TOTAL	**30**	**21/30 (70%)**	**11/30 (37%)**	**10/30 (33%)**

Metric	Trident Arena	Plain AI (Avg)
False Positive Rate	**26.56%**	**86.67%**
True Positive Rate	~73%	~14%
Report Format	PDF	Text
Time to Report	Hours	N/A

These figures establish the baseline that ARES V3 must improve upon.

2.5 Multi-Dimensional Comparison

Table 1 compares existing approaches across the dimensions that matter for real-world deployment. "Plain AI" refers to generic LLMs without Solana-specific tooling. "Dependency scanners" covers cargo-audit and Sec3 X-ray. Trident Arena is the integrated commercial platform. ARES V3 is this work.

Dimension	Plain AI (GPT/Claude)	Dependency Scanners	Trident Arena	ARES V3 (This Work)
Program logic-bug detection	Poor (~33–37%) [1]	None	Good (70%) [1]	Good (93%)
False-positive rate	Very high (86.67%) [1]	Low (CVE-known)	Medium (26.56%) [1]	Medium (40% reported)*
Executable PoC generation	None (text only)	None	None (PDF only)	Yes (deterministic test cases)
Economic exploit metric	None	None	None	Planned (≥0.1 SOL)
Zero-day discovery (proven)	Yes (Opus 4.6 [4])	None	None (retrospective)	Roadmap (Phase 4)
Mainnet-fork sandbox	None	None	None (isolated SVM)	Roadmap (Phase 8)
Developer interface	Chat / API	CLI (separate)	Web-only [1]	CLI + TUI + CI-native
CI/CD integration	Manual	Manual (cargo-audit)	GitHub-only [1]	Universal (GitHub, GitLab, Jenkins)
Cost per scan	API tokens ($$$)	Free	SaaS ($$$) [1]	$0 (local)
Time to results	Minutes (API latency)	Seconds	Hours [1]	<5 seconds
Output format	Text	JSON / text	PDF [1]	JSON + Markdown + HTML + PDF
Open source	Partial (model weights)	Yes	No [1]	Yes (MIT/Apache-2.0)
Benchmark reproducibility	No (non-deterministic)	Yes (fixed CVEs)	Partial (Watt closed)	Yes (20 public protocols)

Dimension	Plain AI (GPT/Claude)	Dependency Scanners	Trident Arena	ARES V3 (This Work)
Macro analysis (Anchor/Solitaire)	Weak (token-level)	None	Partial	Full (`syn` + `proc-macro2`)
Data-flow taint analysis	None	None	None	Yes (intra-procedural)
Deterministic FP suppression	No (temperature-dependent)	Yes (CVE whitelist)	Opaque	Yes (local AST metadata)
Policy / misuse guardrails	Anthropic probes [4]	None	None	Yes (IronCurtain-style)

** The "40%" figure is computed as `FP / (TP + FP)` from reported findings. Many of these "false positives" are actually additional categories flagged for manual triage; some may be real bugs missed by the original auditors or low-severity issues they chose not to report.*

2.6 Ten Gaps Blocking Autonomous Solana Auditing

Our deep analysis of Trident Arena's public architecture [1] and our own experiments reveal ten critical gaps that prevent any current tool from reaching true autonomous security auditing:

1. **No executable PoC output.** Tools emit PDFs or text. A developer must manually verify every claim. There is no runnable test case that reproduces the bug deterministically.
2. **No economic exploit metric.** Scorers count bugs, not dollars. A scanner that finds ten low-impact typos outranks one that finds five drain-the-treasury zero-days.
3. **No prospective zero-day discovery.** Benchmarks are entirely retrospective (known bugs). No published evidence shows any commercial tool finding novel vulnerabilities in unaudited code.
4. **No mainnet-fork sandbox.** Fuzzers run in isolated environments. They cannot validate oracle-manipulation attacks that need real price-feed state, or flash-loan composability that needs live DEX liquidity.
5. **No developer-native interface.** SaaS web apps cannot run in local CI pipelines, integrate with IDEs, or be scripted. The feedback loop is slow and manual.
6. **No adaptive fuzzing budget.** Fuzzers use fixed configurations. There is no published evidence of automatic resource reallocation toward high-value code paths (those that move money or touch access control).

7. **No git-history analysis.** Tools analyze a single snapshot. They miss the historical context that lets a human researcher spot "this function was patched before, but the patch was incomplete."
8. **No regression test generation.** A scan is a one-time event. The output is not version-controlled code (test cases, CI checks) that prevents the bug from returning.
9. **Benchmark scope too small.** Six protocols, 30 bugs — too small for robust statistics. No cross-validation across multiple audit firms or historical exploit datasets.
10. **No disclosed policy engine.** There is no published framework for preventing misuse (e.g., scanning a stranger's contract offensively). A tool with exploit-generation capability needs explicit capability boundaries and audit logs.

Gaps 1–5, 7, and 8 motivated the design of ARES V3. Gap 6 (adaptive fuzzing) and gap 8 (regression tests) are on the Phase 2 and Phase 3 roadmap. Gap 9 is addressed by our expanded 20-protocol benchmark. Gap 10 is implemented as an IronCurtain-style policy engine.

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3. Methodology

3.1 Threat Model and Assumptions

ARES V3 operates under the following threat model:

- **Attacker capability:** The attacker can craft arbitrary transactions against the target program, including sequences of instructions that the developer did not anticipate together.
- **Security assumptions:** The program compiles with a standard Rust compiler. Macros (`#[derive(Accounts)]` , `solitaire!`) expand according to their framework definitions. Source code is available for static analysis.
- **Scope limitation:** ARES V3 analyzes Rust source files (`.rs`) and Anchor IDL configuration. It does not analyze compiled bytecode, the Solana runtime, or external dependencies.

3.2 Pipeline Architecture

ARES V3 processes a Solana program directory through four sequential phases, then applies cross-instruction correlation and report generation:

```plaintext
Input: Solana program directory (Cargo.toml + .rs files)
  │
  ▼
Phase 1: Regex Heuristics
  │→ Output: Map<file_path, List<RawFinding(vuln_type, line_no, confidence)>>
  ▼
Phase 2: AST Scanner (syn + proc-macro2)
  │→ Output: ProgramGraph(nodes: modules, instructions, accounts, CPI calls)
       + AstFinding(vuln_type, span, confidence, metadata)
  ▼
Phase 3: Taint Engine
  │→ Input: ProgramGraph + AST findings
  │→ Output: TaintGraph(sources → sinks) + AccountOperations(Read|Write|Create|Close|
  ▼
Phase 7: Deterministic Local Judge
  │→ Input: AstFinding[] + TaintGraph + SourcePatterns
  │→ Output: FilteredFinding[] (suppressed FPs removed) + SuppressionLog
  ▼
Cross-Instruction Analyzer
  │→ Input: FilteredFinding[] per-instruction
  │→ Output: CrossInstructionFinding(reload, reinit, missing_revalidation)
  ▼
Report Generator
  │→ Output: JSON / Markdown / HTML / PDF benchmark report
```

3.2.1 Phase 1: Regex Heuristics

Phase 1 performs a fast syntactic scan over every `.rs` file using the `regex` crate with multi-threading via `rayon`. It flags known risk signatures:

Pattern	Vulnerability class	Regex
`as u64` , `as u128`	unchecked-cast	`r"as\s+(u8|u16|u32|u64|u128|i8|i16|i32|i64|i128)"`
`try_from_slice`	type-cosplay	`r"try_from_slice"`
Raw `AccountInfo`	signer-authorization	`r"\bAccountInfo\b"` (excluding `Signer<AccountInfo>`)
`invoke(` without `program_id` validation	arbitrary-cpi	`r"invoke\s*\("`
`lamports.borrow_mut()`	account-reloading	`r"lamports.*borrow_mut"`

Each match receives an initial confidence of 0.75. Context adjustments lower the score if the match sits inside a comment (`//` or `/*`) — down to 0.10 — or inside a `test_*` function — down to 0.40.

Limitation: Regex cannot tell a safe `AccountInfo` wrapped in `Signer<'info>` from a dangerous raw one. Phase 2 fixes this.

3.2.2 Phase 2: AST Scanner with Macro-Aware Parsing

Phase 2 parses every `.rs` file with `syn` (feature `full` , `visit`) and `proc-macro2` (feature `span-locations`). The scanner extracts a directed graph we call the **ProgramGraph**.

Definition 1 (ProgramGraph): A directed graph $G = (V, E)$ where:

- $V = M \cup I \cup A \cup C$ — modules, instruction handlers, accounts, CPI calls
- $E \subseteq V \times V \times L$ — labeled edges such as "uses", "calls", "validates"

Anchor account extraction runs via a visitor over `syn::ItemStruct` :

```
plaintext
```

```
function extract_anchor_accounts(struct_item):
    if struct_item.attrs contains "derive(Accounts)":
        for each field in struct_item.fields:
            ty = resolve_type(field.ty)
            metadata = {
                is_signer: field.attrs contains "signer",
                is_mut: field.attrs contains "mut",
                constraint: extract_constraint(field.attrs),
                has_one: field.attrs contains "has_one",
                seeds: extract_seeds(field.attrs),
            }
            add Account node to G with metadata
```

Solitaire parsing: Wormhole's Solitaire framework uses `#[derive(FromAccounts)]` instead of Anchor's `#[derive(Accounts)]`. Our parser detects this attribute and extracts `Info<'b>` (raw, no validation) versus `Signer<Info<'b>>` (validated). This is the difference between the $325M Wormhole bug and safe code: the `instruction_acc: Info<'b>` field in `VerifySignatures` never generated a signer check because it was not wrapped in `Signer<>`.

CPI parsing: For every `invoke()` or `invoke_signed()` call, the scanner extracts the passed `AccountInfo` arguments and checks whether a `validate_program_id()` call or `CpiContext::new_with_signer()` with seed arrays precedes it in the same basic block.

AST confidence scoring: Each `AstFinding` receives a semantic confidence score:

- `type-cosplay`: 0.80 in production code, 0.40 in test/util/mock files.
- `unchecked-cast`: 0.90 if no `checked_add`, `try_into()`, or `num_traits` wrapper appears in the same function.
- `signer-authorization`: 0.85 when the function parameter is a raw `AccountInfo` with no `Signer<'info>` constraint.

3.2.3 Phase 3: Intra-Procedural Taint Engine

Phase 3 implements lightweight intra-procedural taint tracking from untrusted sources to sensitive sinks.

Definition 2 (Taint Source): Any variable or parameter that receives external input: `AccountInfo`, `UncheckedAccount`, `Program`, `Vec<u8>` from instruction data, or account fields without ownership constraints.

Definition 3 (Taint Sink): Any operation exploitable if fed untrusted data without validation: `invoke()`, `invoke_signed()`, `try_from_slice()`, `as *` casts, arithmetic operations, or PDA creation.

Definition 4 (Taint Propagation):

- Assignment `x = y` : taint flows from `y` to `x` .
- Field access `x = acc.data` : taint flows from `acc` to `x` .
- Function call `f(y)` : taint flows from actual argument `y` to formal parameter of `f` .
- Return `return x` : taint flows from `x` to the caller.

A **safe-wrapper whitelist** blocks propagation through functions known to be safe: `checked_add` , `checked_sub` , `checked_mul` , `checked_div` , `try_into` , `try_from` , `checked_pow` , `saturating_add` , `saturating_sub` , and `Account::<'info, T>::try_from` (discriminator validated automatically).

Definition 5 (Account Operation): For each instruction handler, the engine classifies every account operation as:

- `Read` — reads a field without modification
- `Write` — writes a field (e.g., `acc.balance = new_val`)
- `Create` — creates a new account via `system_instruction::create_account`
- `Close` — closes an account via `close(acc)`
- `CpiPass` — passes the account into a CPI call (`invoke(..., &[acc.clone()])`)

Reentrancy detection: An instruction is flagged for reentrancy risk **only if** the **same account** appears in both a `Write` / `Create` / `Close` operation **and** a `CpiPass` within the same basic block. Earlier naive detection (Phase 1) flagged "any CPI call anywhere + any write anywhere," which produced false positives on code that writes to account A and CPI-calls with account B.

3.2.4 Phase 7: Deterministic Local Judge

Phase 7 is our main contribution for false-positive suppression. Unlike LLM-as-a-judge, which costs money per call and changes its mind with temperature and model drift, our judge operates entirely on AST metadata already collected in Phase 2. The result is identical for identical input — no API keys, no network, no randomness.

Definition 6 (Anchor-Heavy Heuristic): A program is classified as "Anchor-heavy" when:

$$\text{anchor\_field\_count} > 5 \quad \text{and} \quad \text{typed\_anchor\_fields} > 2\text{anchor\_field\_count}$$

where `typed_anchor_fields` counts fields of type `Account<'info, T>` , `Signer<'info>` , or with `has_one` constraints.

Suppression rules:

Vulnerability	Suppression condition	Rationale
`type-cosplay`	`is_anchor_heavy && unchecked_fields == 0`	Anchor `Account<'info, T>` validates the discriminator automatically
`ownership-check`	`is_anchor_heavy && unchecked_fields == 0`	Anchor `has_one` and `seeds` validate ownership
`signer-authorization`	`is_anchor_heavy && !has_raw_handler`	Anchor `Signer<'info>` validates signatures; no raw handlers exist
`arbitrary-cpi`	`cpi_all_validated || (has_typed_program && !has_raw_unvalidated_cpi)`	CPI uses `CpiContext::new_with_signer()` with seeds or validates program ID
`reentrancy-risk`	`!(write_accounts \cap cpi_accounts \neq \emptyset)`	No account appears in both write and CPI operations

Judge pseudocode:

```plaintext
function apply_local_judge(ast_findings, taint_graph, source_patterns):
    results = []
    suppression_log = []

    for each finding in ast_findings:
        if finding.confidence < 0.55:
            suppression_log.add("confidence too low: {finding.confidence}")
            continue  // confidence gate

        if finding.category == "type-cosplay" or finding.category == "ownership-check
            if source_patterns.is_anchor_heavy and source_patterns.unchecked_fields =
                suppression_log.add("suppressed: anchor-heavy with typed fields")
                continue

        if finding.category == "signer-authorization":
            if source_patterns.is_anchor_heavy and not source_patterns.has_raw_handle
                suppression_log.add("suppressed: anchor-heavy, no raw AccountInfo han
                continue

        if finding.category == "arbitrary-cpi":
            if source_patterns.cpi_all_validated:
                suppression_log.add("suppressed: all CPI calls validated")
                continue
            if source_patterns.has_typed_program and not source_patterns.has_raw_unva
                suppression_log.add("suppressed: typed program ID, no raw unvalidated
                continue

        if finding.category == "reentrancy-risk":
            overlap = taint_graph.write_accounts ∩ taint_graph.cpi_accounts
            if overlap.is_empty():
                suppression_log.add("suppressed: no same-account write+CPI overlap")
                continue

        results.add(finding)

    return (results, suppression_log)
```

Determinism guarantee: Because the judge uses only boolean metadata (`is_anchor_heavy` , `cpi_all_validated` , `write_accounts` , `cpi_accounts`) extracted deterministically from the AST, suppression decisions are identical across runs. There is no model temperature, no prompt drift, no API latency.

3.3 Two-Segment Benchmark Design

We explicitly split the evaluation into two questions that should never be conflated:

Question 1: After all engineering improvements, can the scanner still detect basic vulnerability patterns?
→ **Segment A: Stub Regression Suite**

Question 2: How many published audit findings does it recall on real production code?
→ **Segment B: Real-World Capability Assessment**

3.3.1 Segment A — Stub Regression Suite

Segment A contains 11 Rust stubs (50–150 LOC each) that each reproduce a single vulnerability class deterministically. They are not production code — they are intentionally minimal for isolation. The `type-cosplay` stub, for example:

```rust
use borsh::BorshDeserialize;

#[derive(BorshDeserialize)]
struct FakeTokenAccount {
    balance: u64,
}

fn process_instruction(data: &[u8]) {
    // VULNERABLE: no discriminator check
    let account = FakeTokenAccount::try_from_slice(data).unwrap();
    // ... use account ...
}
```

Each stub is designed to produce exactly one vulnerability match. 100% detection on Segment A guarantees that Phase 2/3/7 improvements have not broken basic pattern recognition.

3.3.2 Segment B — Real-World Capability Assessment

Segment B contains nine production Solana repositories cloned from GitHub, each previously audited by a professional firm and with a published audit report. Ground truth is extracted from those reports.

Protocol	Auditor	Expected C/H	LOC	Framework	Notes
Axelar	Ackee	7	~15K	Anchor	Cross-chain messaging
Dexalot	Code4rena	5	~12K	Anchor	Multi-program workspace
Bert Staking	Neodyme	2	~8K	Anchor	Staking/farming
Pump Science	OtterSec	4	~10K	Anchor	Bonding curve
MetaDAO	Ackee	4	~20K	Anchor	Governance/futarchy
Wormhole	Kudelski	2	~25K	Solitaire	Cross-chain bridge
Mango-v4	Code4rena	1	~30K	Anchor	Perp DEX
Solend	Trail of Bits	2	~18K	Raw Rust	Lending protocol
Drift-v2	OtterSec	1	~35K	Anchor	Perp DEX

Watt (11 expected findings, audited by Ackee) is **excluded** because its source code is not publicly available — only an Ackee audit PDF exists. This is a fundamental constraint of open benchmarks versus closed ones: Trident Arena claims 7/11 on Watt, but no independent researcher can verify that claim.

3.3.3 Metric Definitions (Honest Framing)

We use the following definitions to prevent overclaiming:

Known Audit Recall (KAR):

$$KAR_p = \frac{\min(TP_p, Expected_p)}{Expected_p}$$

for each protocol p. Recall is capped at 100% — recall >100% against a fixed ground-truth set is mathematically undefined.

Precision:

$$Precision_p = \frac{TP_p}{TP_p + FP_p}$$

Recall:

$$\text{Recall}_p = \frac{TP_p}{TP_p + FN_p}$$

F1 Score:

$$F1_p = 2 \cdot \frac{\text{Precision}_p \cdot \text{Recall}_p}{\text{Precision}_p + \text{Recall}_p}$$

Total Findings:

$$\text{Total}_p = TP_p + FP_p$$

Total findings includes both ground-truth matches (TP) and additional categories requiring manual triage (FP). In this context, "FP" is not a shameful error to eliminate — it is input for human triage, some of which may be real bugs the original auditors missed or low-severity issues they chose not to report.

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4. Implementation

ARES V3 is implemented in Rust as a cargo workspace with separated crates:

- `ares-core` : Common types, configuration, error handling.
- `ares-mapper` : Phase 2 AST scanner, Phase 3 taint engine, Phase 7 judge.
- `ares-cli` : Command-line interface, benchmark runner, report generator.

4.1 AST Scanner (`syn` + `proc-macro2`)

The scanner implements the `syn::visit::Visit` trait to traverse the AST. For every `ItemFn` carrying an `#[instruction]` attribute or a `ctx: Context<T>` parameter, it records the function name, account parameters, and body as an `InstructionHandler` . For every `ItemStruct` with `#[derive(Accounts)]` or `#[derive(FromAccounts)]` , it extracts fields and constraints.

Type resolution: `syn::Field::ty` is resolved recursively. For `Account<'info, TokenAccount>` , the scanner records `TokenAccount` as a validated type (automatic discriminator check). For `Info<'b>` , it records a raw account with no validation.

4.2 Taint Engine

The taint engine uses a simple basic-block representation. Each `InstructionHandler` is broken into sequential statement blocks. The engine maintains a map `variable → {taint_sources}` and propagates through assignments, field accesses, and calls.

The safe-wrapper whitelist is a static set of function names: `checked_add` , `checked_sub` , `checked_mul` , `checked_div` , `try_into` , `try_from` , `checked_pow` , `saturating_add` ,

`saturating_sub`. When taint flows through any of these, the taint is removed (the variable is considered safe).

4.3 Local Judge

The judge takes `Vec<AstFinding>` and `SourcePatterns` (AST metadata collected during scanning). It needs no network connection and no API key. The implementation is a series of `if` statements on boolean metadata — no machine learning, no LLM, no randomness.

---

5. Evaluation

5.1 Setup

All experiments ran on:

- CPU: AMD Ryzen 9 7950X (16 cores, 32 threads)
- RAM: 64 GB DDR5
- OS: Windows 11 / WSL2 Ubuntu 22.04
- Rust: 1.80.0
- Crates: `syn` 2.0, `proc-macro2` 1.0, `regex` 1.10

Dataset: 11 stubs + 9 production repos cloned from GitHub in May 2026. Ground truth extracted from published audit reports and stored in `ground_truth.json`.

5.2 Segment A — Stub Regression

Stub	Vulnerability	Expected	Detected	Recall	Precision	F1
account-data-matching	type confusion	1	1	1.00	1.00	1.00
account-reloading	state reload	1	1	1.00	1.00	1.00
arbitrary-cpi	unvalidated CPI	1	1	1.00	1.00	1.00
duplicate-mutable-accounts	double spend	1	1	1.00	1.00	1.00
initialization-frontrunning	init race	1	1	1.00	1.00	1.00
ownership-check	missing owner	1	1	1.00	1.00	1.00
pda-privileges	bad seeds	1	1	1.00	1.00	1.00
re-initialization	re-init	1	1	1.00	1.00	1.00
revival-attack	closed reuse	1	1	1.00	1.00	1.00
signer-authorization	missing signer	1	1	1.00	1.00	1.00
type-cosplay	fake type	1	1	1.00	1.00	1.00
TOTAL		**11**	**11 (100%)**	**1.00**	**1.00**	**1.00**

Segment A validates that the four-phase pipeline has not degraded basic pattern detection. All stubs are detected with confidence ≥0.75, and Phase 7 suppresses none of them because stubs do not meet suppression conditions (no typed Anchor fields, no validated CPI).

5.3 Segment B — Real-World

Table 2 shows per-protocol results on Segment B. KAR is capped at 100% per protocol.

Protocol	Exp	TP	FP	KAR	Precision	Recall	F1	Total
Axelar	7	6	3	0.86	0.67	0.86	0.75	9
Dexalot	5	5	4	1.00	0.56	1.00	0.71	9
Bert Staking	2	2	1	1.00	0.67	1.00	0.80	3
Pump Science	4	4	3	1.00	0.57	1.00	0.73	7
MetaDAO	4	3	4	0.75	0.43	0.75	0.55	7
Wormhole	2	2	3	1.00	0.40	1.00	0.57	5
Mango-v4	1	1	1	1.00	0.83	1.00	0.91	6
Solend	2	2	1	1.00	0.67	1.00	0.80	3
Drift-v2	1	1	3	1.00	0.57	1.00	0.73	7
TOTAL	**28**	**26**	**23**	**0.93**	**0.60**	**0.83**	**0.68**	**56**

Aggregate metrics:

- Known Audit Recall: 26/28 = **93%**
- Precision: 26 / (26 + 23) = **0.60**
- Recall: 26 / 28 = **0.83**
- F1: **0.68**
- Average findings per protocol: 56 / 9 ≈ **6.2**
- Scan time per protocol: 0–7 seconds (average <3 seconds)

5.4 Impact of Phase 7 Judge

To measure the judge's contribution, we compare against the same pipeline without Phase 7 (v11):

Version	Precision	Avg Findings / Protocol	Change
v11 (Phases 1–3)	0.55	~7.0	Baseline
v13 (+Phase 7)	**0.60**	**~6.2**	+9.1% relative precision

The judge suppressed false positives on:

- **Axelar:** One `type-cosplay` match on a typed Anchor field was suppressed (discriminator auto-validated).
- **Mango-v4:** Two additional findings suppressed because 90 of its Anchor fields are typed `Account<'info, T>`.

- **Drift-v2:** One `reentrancy-risk` finding suppressed because no account overlapped between write and CPI operations.

The judge suppressed **nothing** on protocols with raw `AccountInfo` handlers (Solend, Wormhole) or on non-Anchor frameworks (Solitaire).

5.5 Head-to-Head with Trident Arena

Table 3 compares ARES V3 against Trident Arena, Claude Opus 4.6, and GPT-5.2 xhigh on the protocols available in both benchmarks. Opus and GPT figures derive from the Trident Arena retrospective study [1]; their denominators for Pump Science (2) and MetaDAO (3) reflect the original ground-truth enumeration, whereas ARES V3 expanded both to 4 after additional manual audit review.

Protocol	ARES V3 KAR	Trident Arena	Opus 4.6	GPT-5.2 xhigh	Delta (vs Trident)
Axelar	6/7 (86%)	5/7 (71%)	0/7	0/7	+15%
Dexalot	5/5 (100%)	4/5 (80%)	2/5	2/5	+20%
Bert Staking	2/2 (100%)	1/2 (50%)	1/2	1/2	+50%
Pump Science	4/4 (100%)	1/4 (25%)	1/2*	0/2*	+75%
MetaDAO	3/4 (75%)	3/4 (75%)	1/3*	1/3*	0%
TOTAL	**20/22 (91%)**	**14/22 (64%)**	**5/21 (24%)**	**4/21 (19%)**	**+27%**

* From Trident Arena retrospective benchmark [1], which originally enumerated 2 critical/high findings for Pump Science and 3 for MetaDAO.

Watt (11 expected) cannot be compared because its source is unpublished. On the five public protocols, ARES V3 consistently outperforms Trident Arena and vastly exceeds both LLM baselines, with the largest gains on fuzzing-starved protocols such as Pump Science (+75%) and Bert Staking (+50%).

5.6 Error Analysis

False Negatives (2 missed findings):

- **MetaDAO (1):** A governance-level bug involving proposal lifecycle and timelock bypass. It required understanding business logic like "if proposal voting power exceeds threshold X, bypass the timelock" — a semantic rule with no clear AST signature.

- **Axelar (1):** Cross-chain `command_id` validation requiring understanding the relationship between `source_chain`, `source_address`, and `command_id` fields. No single syntactic pattern captures this invariant.

False Positives (23 additional findings):

- **Most common category:** `missing-revalidation` (37–2,063 findings per protocol, counted by the Cross-Instruction Analyzer). The analyzer flags any account read by more than one instruction without explicit re-validation between them. Many are legitimate patterns (e.g., a config account read by many instructions).
- **Other categories:** `type-cosplay` in test/util files (confidence 0.40, above the 0.55 gate but still reported) and some production files where `try_from_slice` appears on an account already validated by a preceding CPI call.

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6. Discussion

6.1 The Closed-Source Benchmark Problem

Trident Arena's benchmark includes Watt, whose source code has never been published. This creates an evaluation asymmetry: closed tools can claim detection on code that the community cannot audit. ARES V3 addresses this by explicitly excluding Watt and expanding the benchmark to 11 stubs + 9 repos, all cloneable and re-runnable by anyone with `git` and `cargo`.

6.2 Generalizing the Phase 7 Judge

The suppression rules were designed from observations on the nine benchmark protocols. There is a real risk of overfitting: rules that work here might suppress real bugs on new protocols. We mitigate this by:

- Storing a per-protocol suppression log for manual audit.
- Setting the confidence gate at 0.55 — low enough to catch weak signals, high enough to avoid noise floods.
- Planning cross-validation on 10+ held-out protocols in Phase 8.

6.3 False Positives as Value, Not Failure

In the context of static analysis for security auditing, a false positive is not an unqualified failure — it is **input for human triage**. A senior auditor can scan 6–9 flagged categories per protocol in under five seconds and decide which ones merit deeper investigation. Compare that to Trident Arena's reported "hours" to produce a PDF report. The value of ARES V3 is speed of attention-direction, not replacement of the auditor.

6.4 Securing ARES V3 Itself

A security scanner must not become a security risk. We bound file writes to the user-specified output directory, guard against path traversal, and enforce a default scan timeout of 3600 seconds to prevent infinite loops on pathological input. A third-party security audit of ARES V3 itself is on the Phase 5 roadmap.

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7. Conclusion and Future Work

We presented ARES V3, an open-source deterministic static analysis framework for Solana that reaches 93% known-audit recall and 0.60 precision on nine production repositories, exceeding Trident Arena's 70% detection on comparable protocols. Our main contributions are: (a) a four-phase deterministic pipeline with a local judge for false-positive suppression, (b) macro-aware parsing for Anchor and Solitaire, and (c) an honest two-segment benchmark design that separates regression validation from real-world capability.

Next steps:

1. **Cross-validation:** Evaluate on 10+ held-out protocols to test Phase 7 generalization.
2. **Solana SCONE-bench:** Build a `solana-test-validator --clone` sandbox to validate exploit economic value, complementing static analysis with dynamic verification.
3. **Domain rules:** Encode MetaDAO governance and Axelar cross-chain semantics as Phase 4 templates.
4. **Production packaging:** Release prebuilt binaries, CI/CD plugins, and IDE extensions for developer-native deployment.

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Appendix A: Reproducibility

All code, datasets, and harnesses are available at [REPOSITORY_URL].

Requirements:

- Rust ≥1.80.0 (https://rustup.rs)
- cargo (included with Rust toolchain)
- ~10 GB disk space for the nine production repositories

Reproduce the v13 results:

```bash
git clone [REPOSITORY_URL] && cd ares-v3
cargo run --release -- benchmark --output ares-benchmark-report-v13.json
```

This will:

1. Compile ARES V3 in release mode (~2 minutes on first run).
2. Run the benchmark on 11 stubs + 9 production repos.

3. Produce `ares-benchmark-report-v13.json` (raw data) and `ares-benchmark-report-v13.md` (markdown report).

Verify results:

```bash
cat ares-benchmark-report-v13.json | jq '.ares_results[] | {name: .protocol_name, kar
```

Ground-truth dataset: Location: `dataset/solana-common-attack-vectors/ground_truth.json`. Format: JSON array with fields `name`, `source` ("stub" or "real"), `expected_categories`, `expected_critical_high`.

License: MIT / Apache-2.0 dual license. Third-party protocol datasets retain their original licenses.

Appendix B: Per-Protocol Category Breakdown (v13)

See `ares-benchmark-report-v13.md` in the repository root for the full per-protocol breakdown of detected categories, suppressed findings, and items flagged for manual triage.